Logos Reference Manual

A Logic for Interpreted and Verified AI Reasoning

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— January 31, 2026 —

Primary References:

"The Construction of Possible Worlds", Brast-McKie, (under review), 2025.

"Counterfactual Worlds", Brast-McKie, *J. Phil. Logic*, 2025.

"Identity and Aboutness", Brast-McKie, *J. Phil. Logic*, 2021.

Abstract

This reference manual provides the formal specification of the Logos which is an extensible formal language of thought designed to train AI systems in verified logical reasoning. The Logos is interpreted by a hyperintensional task semantics designed to support operators for reasoning with tense, modal, counterfactual conditional, constitutive explanatory, causal, and many other types of expressive resources. The semantics and proof theory for the foundational layers of the Logos are then successively extended by a plugin system to support a wide range of additional operators for complex reasoning to plan and evaluate actions in multi-agent systems under conditions of uncertainty.

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1 Introduction

This reference manual provides the formal specification for the Logos proof system, semantics, metalogic, and library of theorems. The semantics is organized around the *semantic frames* that provide the primitive structure used to interpret the logical operators included in the Logos. By successively extending the primitive resources included in a frame, it becomes possible to provide semantic clauses for a new range of operators, thereby extending the expressive power of the Logos for systematic reasoning.

The Logos is interpreted over a complete lattice called a *frame* which provides the semantic primitives for the *Constitutive Foundation* of the theory. Whereas intensional semantic theories take total instantaneous states of a system to be primitive, the state semantics takes states and their parthood relations to be primitive to provide fine-grained semantic contents. These fine-grained contents can then be used to interpret standard first-order resources along with constitutive explanatory operators for *essence*, *ground*, *reduction*, and *propositional identity* which formalize 'sufficient for', 'necessary for', 'what it is for', and 'just is for' explanatory relationships.

The *Dynamical Foundation* extends the primitive semantic resources given by a frame to include a *task relation* that is parameterized by *times* to govern the lawful transitions by which the states in a system evolve. These elements are used to construct *world sates* which represent total instantaneous states of a system, as well as *world histories* which model the possible evolutions of that system by mapping durations to world states in lawful. For instance, in modeling a game of chess, the world states correspond to the legal complete configurations of the chessboard, and the world histories correspond to the possible games of chess. World histories are used to provide semantic clauses for modal, temporal, counterfactual, and causal operators in addition to those included in the *Constitutive Foundation*.

A world history and time suffice to assign truth-values to all sentences of the foundational layers of the Logos for planning and evaluating actions. Plugins extend the expressive power of the foundational layers of the Logos by adding semantic primitives as well as new contextual parameters at which sentences are evaluated. For instance, given a *credence function* which assigns likelihoods to propositions, a *belief parameter* is required along with a world history and time to provide a semantics for the belief operator. Given a world history, time, and belief parameter, a proposition is believed just in case the credence in that proposition is greater than the belief parameter included in the point of evaluation.

1.1 Extension Dependencies

This section presents the layer extensions by which the Logos is constructed where Figure 1 shows the dependency structure among extensions. The first two layers provide an essential foundation followed by a modular plugin system for language extensions and an additional agential layer for multi-agent, self-reflective, and social reasoning.

Whereas the *Constitutive Foundation* and *Dynamical Foundation* form the required base, the middle extensions (*Epistemic*, *Abilities*, *Normative*, *Choice*, *Spatial*, and others) are modular plugins that can be combined in any subset. The *Agential Extension*, *Social Extension*, and *Reflection Extension* all inherit from the layers above and are free to interact with each other. The *Agential Extension* provides third-person multi-agent reasoning, the *Social Extension* provides the construction of the common ground and trust relations, and the *Reflection Extension* enables first-person metacognition.

1.2 Layer Descriptions

1. **Constitutive Foundation**: Provides the hyperintensional base with predicates (F, G), functions (f, g), variables (x, y), booleans ($\neg$, $\wedge$, $\vee$), extremal ($\bot$, $\top$), lambda abstraction (λ), identity ($=$), and constitutive operators (propositional identity $\equiv$, grounding $\leqslant$, essence $\sqsubseteq$, reduction $\Rightarrow$) over a complete lattice with bilateral propositions (verifier/falsifier pairs). This layer enables fine-grained distinctions between propositions with identical truth-values and modal profiles but different verification conditions, essential for exact specification of what makes a claim true.
2. **Dynamical Foundation**: Adds quantifiers ($\forall$, $\exists$), modals ($\Box$, $\Diamond$) for necessity and possibility, tense (H, G, P, F, $\lhd$, $\rhd$) for reasoning across time, store/recall operators ($\downarrow$, $\uparrow$) for temporal reference, actuality (Act) for distinguishing actual from possible, counterfactual conditionals ($\Box\!\!\rightarrow$, $\Diamond\!\!\rightarrow$) for hypothetical reasoning, and causal operators ($\bigcirc\!\!\rightarrow$, $\odot\!\!\rightarrow$) for explanatory relations over a duration. Together these resources enable reasoning about action sequences, future contingency, what would have happened under different circumstances, and causal relationships.
3. **Epistemic Extension**: Introduces epistemic modals (Mi, Mu), belief (B), probability (Pr), and indicative conditional operators ($\hookrightarrow$) for reasoning under uncertainty. By extending the semantic framework with credence functions to assign probabilities to states and chance functions which assign probabilities to state transitions, agents can track how evidence updates beliefs as well as engaging in suppositional reasoning.

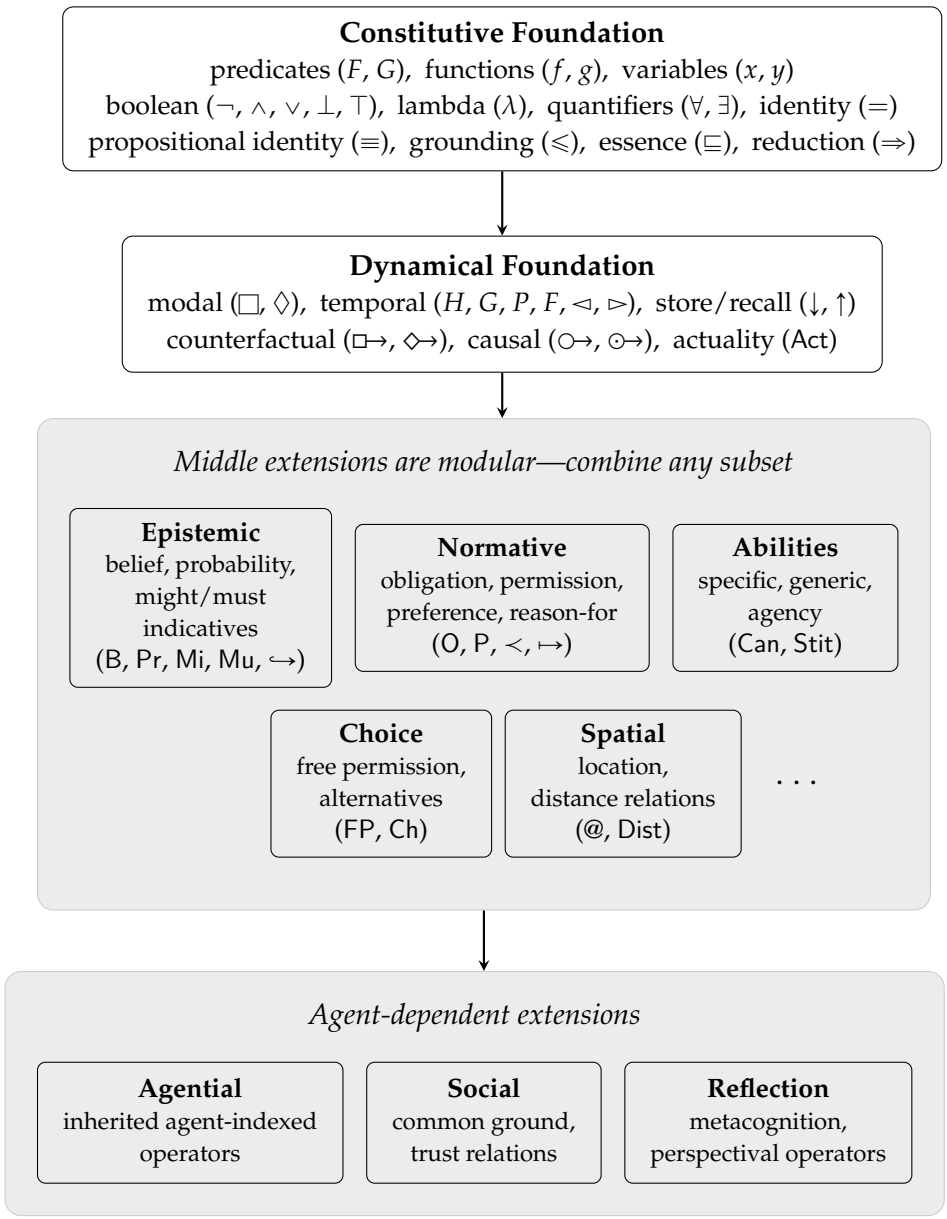

Figure 1: Extension dependency structure in the Logos system.

4. **Abilities Extension**: Provides ability (Can) and STIT (stit) operators for reasoning about capacities. Ability modals express what an agent can bring about through their capacities, with stronger truth conditions than circumstantial possibility since ability entails possibility but not *vice versa*. This enables reasoning about what agents are capable of accomplishing given their intrinsic capacities and circumstances.
5. **Normative Extension**: Adds obligation (O), permission (P), along with a preference operator ($<$) with value orderings over states for reasoning about what ought to be done, what is permitted, and how to rank alternatives. This enables ethical reasoning about alternatives.
6. **Choice Extension**: Introduces free choice permission (FP) and best choice (Ch) operators for reasoning about permission over alternatives, enabling reasoning about which alternatives are permitted and how permissions combine and distribute.
7. **Spatial Extension**: Provides location-dependent operators and spatial relations for reasoning about where things are and how they are oriented relative to an orientation basis. This supports navigation, containment reasoning, and location-sensitive planning.
8. **Agential Extension**: Indexes epistemic and normative operators to specific agents (B_a, $<$, O_a, ...), enabling reasoning about what different agents believe, prefer, and are obligated to do. This transforms single-perspective reasoning into multi-agent coordination where agents can model each other's epistemic and normative states.
9. **Social Extension**: Provides resources for the construction of a common ground and group-level reasoning, building on the *Agential Extension* to enable reasoning about shared beliefs, building trust relations between agents, and coordinate on common goals.
10. **Reflection Extension**: Adds first-person metacognitive operators for self-directed reasoning about one's own beliefs, abilities, and progress towards goals. This enables an agent to reason about what it is in a good position to believe given an evaluation of its own limitations and track relative to other agents. Essential for AI systems that must understand their own capabilities and benefit from the capacity to learn by appealing to the testimony of others with trusted expertise.

For the Lean implementations corresponding to each layer, consult the `Theories/Logos/SubTheories/` directory, where foundational components are in `Foundation/`, dynamical extensions in `Dynamical/`, and the various logical extensions in their respective subdirectories.

1.3 Document Organization

This reference manual is organized as follows:

Section 2 presents the *Constitutive Foundation*, including the mereological state space, verification and falsification clauses, and bilateral propositions.

Section 3 introduces the syntactic primitives of the *Dynamical Foundation*, including modal, temporal, and counterfactual operators and provides the semantic framework, including task frames, world-histories, and truth conditions.

Section 4 extends the *Dynamical Foundation* with structures for belief, knowledge, probability, and indicative conditionals for reasoning under uncertainty.

Section 5 extends the *Dynamical Foundation* with structures for obligation, permission, and value orderings for ethical reasoning about alternatives.

Section 6 extends the *Dynamical Foundation* with location-dependent operators and spatial relations for reasoning about orientation and navigation.

Section 7 provides multi-agent reasoning structures that index epistemic and normative operators to specific agents, requiring at least one middle extension to be loaded.

Section 8 specifies the Reflection Extension for first-person metacognitive reasoning, including truth conditions, derived operators, and temporal interactions.

1.4 Lean Implementation

The Logos system is implemented in Lean 4 with Mathlib. Each definition in this reference has a corresponding Lean implementation, referenced using the `\leansrc{}{}` command.

For example, `Logos.Foundation.Frame.ConstitutiveFrame` refers to the `ConstitutiveFrame` definition in the `Logos.Foundation.Frame` module. The implementation can be found in the `ProofChecker` repository in the `Theories/Logos/` directory.

2 Constitutive Foundation

The *Constitutive Foundation* provides the foundational semantic structure based on exact truthmaker semantics. Evaluation is hyperintensional, distinguishing propositions that agree on truth-value across all possible worlds but differ in their exact verification and falsification conditions.

2.1 Syntactic Primitives

The Constitutive Foundation interprets the following syntactic primitives:

- **Variables**: $v_1, v_2, v_3, \ldots$ (ranging over states)
- **Individual constants**: $a, b, c, \ldots$ (0-place function symbols)
- **n-place function symbols**: $f, g, h, \ldots$
- **n-place predicates**: $F, G, H, \ldots$
- **Sentence letters**: $p, q, r, \ldots$ (0-place predicates)
- **Lambda abstraction**: $\lambda v.\varphi$ (binding variable v in formula φ)
- **First-order identity**: $=$ (identity between terms)
- **Logical connectives**: $\neg, \wedge, \vee, \top, \bot, \equiv$

Remark 2.1 (Variable Naming Convention). Object-language variables are denoted $v, v_1, v_2, v_3, \ldots$ to distinguish them from metalanguage variables x, y, z which range over durations in the Dynamical Foundation. When a single variable suffices (e.g., in lambda abstractions), we write simply v.

Definition 2.2 (Well-Formed Formulas)**.** The set of *well-formed formulas* is defined inductively where F over n-place predicates, t_i over terms, and v over variables:

$$\varphi, \psi ::= F(t_1, \ldots, t_n) \mid \top \mid \bot \mid \neg\varphi \mid (\varphi \wedge \psi) \mid (\varphi \vee \psi) \mid (\varphi \equiv \psi) \mid (\lambda v.\varphi)(t)$$

See `Logos.SubTheories.Foundation.Syntax.ConstitutiveFormula`.

2.2 Derived Operators

Definition 2.3 (Derived Operators)**.** The following notation conventions define syntactic sugar for the Constitutive Foundation language:

1. **Material Conditional**: $\varphi \to \psi := \neg\varphi \vee \psi$
2. **Biconditional**: $\varphi \leftrightarrow \psi := (\varphi \to \psi) \wedge (\psi \to \varphi)$

First-order quantifiers are *not* primitive in the Constitutive Foundation; first-order quantification is introduced in the Dynamical Foundation layer.

See `Logos.Dynamics.Syntax.CoreFormula`.

2.3 Constitutive Frame

Definition 2.4. A *constitutive frame* is a structure $\mathcal{F} = \langle \mathcal{S}, \sqsubseteq \rangle$ where:

- $\mathcal{S}$ is a nonempty set of *states*
- $\sqsubseteq$ is a partial order on $\mathcal{S}$ where $\langle \mathcal{S}, \sqsubseteq \rangle$ is a complete lattice

Remark 2.5. The constitutive frame is non-modal: neither possibility nor compatibility can be defined since this requires the task relation introduced in the Dynamical Foundation. Taking S itself to consist of *possible* states disrupts closure under fusion makes the conjunction of incompatible propositions empty, stripping the semantics of its capacity to propositions differing in subject-matter with no gain in simplicity or elegance.

Remark 2.6. The lattice structure provides:

- **Null state** $\square$: The bottom element (fusion of the empty set)
- **Full state** $\blacksquare$: The top element (fusion of all states)
- **State Fusion** $s \sqcup t$: The least upper bound of s and t
- **Pointwise Fusion** $(f \sqcup g)(s_1, \ldots, s_n) := f(s_1, \ldots, s_n) \sqcup g(s_1, \ldots, s_n)$

See `Logos.Foundation.Frame.ConstitutiveFrame` for the implementation.

2.4 Constitutive Model

Definition 2.7. A *constitutive model* is a structure $\mathcal{M} = \langle \mathcal{S}, \sqsubseteq, |\cdot| \rangle$ where $\langle \mathcal{S}, \sqsubseteq \rangle$ is a constitutive frame and $|\cdot|$ is an *interpretation function* satisfying:

- **n-place function symbols** $|f| : (\texttt{Fin}\ n \to \mathcal{S}) \to \mathcal{S}$
- **n-place predicates** $|F|^+, |F|^- : ((\texttt{Fin}\ n \to \mathcal{S}) \to \mathcal{S}) \to \mathsf{Prop}$ are the *verifier function* and *falsifier function* types which are required to satisfy:[1]
  - $f \sqcup g : |F|^\pm$ for any $f, g : |F|^\pm$.
  - $\bigsqcup(a_1, \ldots, a_n) \sqsubseteq f(a_1, \ldots, a_n)$ for any $f : |F|^\pm$ and $a_1, \ldots, a_n : \mathcal{S}$.

Remark 2.8. Constants and sentence letters are covered by the trivial cases of 0-place functions and 0-place predicates:

[1] An equivalent formulation takes $|F|^+, |F|^- : \texttt{Set}\ ((\texttt{Fin}\ n \to \mathcal{S}) \to \mathcal{S})$ since in Lean 4, $\texttt{Set}\ \alpha$ is defined as $\alpha \to \mathsf{Prop}$. Whereas the Set notation emphasizes the collective aspect, the Prop notation emphasizes the logical aspect; both capture the same mathematical structure.

- **Constant symbols** $|c| : \mathcal{S}$ is the type for 0-place functions.
- **Sentence letters** $|p|^+, |p|^- : \mathcal{S} \to \mathsf{Prop}$ are the *verifier function type* and *falsifier function type* for 0-place predicates.

See `Logos.Foundation.Model.ConstitutiveModel`.

2.5 Variable Assignment

Definition 2.9. A *variable assignment* $\sigma : \mathrm{Var} \to \mathcal{S}$ maps each variable name to a state in the state space which represents an object.

Notation 2.10. Greek letters ($\tau, \alpha, \beta, \ldots$) are reserved for world-histories. The letter σ (with subscripts $\sigma_1, \sigma_2, \ldots$) denotes variable assignments, chosen for compatibility across LaTeX, markdown, and Lean notation.

Definition 2.11 (Term Algebra)**.** The set Term of *terms* is defined inductively:

$$t ::= v \mid f(t_1, \ldots, t_n)$$

where v ranges over variables, f over n-place function symbols, and c is used to indicate constants 0-place function symbols.

The *free variables* $\mathsf{FV}(t)$ of a term are defined by:

$$\mathsf{FV}(v) = \{v\}$$
$$\mathsf{FV}(f(t_1, \ldots, t_n)) = \bigcup_{i=1}^{n} \mathsf{FV}(t_i)$$

Substitution $t[s/v]$ (replacing v with s in t) is defined by:

$$v'[s/v] = \begin{cases} s & \text{if } v' = v \\ v' & \text{otherwise} \end{cases}$$
$$f(t_1, \ldots, t_n)[s/v] = f(t_1[s/v], \ldots, t_n[s/v])$$

See `Logos.SubTheories.Foundation.Syntax.Term`.

Definition 2.12. The *v-variant* $\sigma[s/v]$ of the variable assignment σ differs at most from σ by assigning v to s.

Definition 2.13. The *extension* of a term given a model $\mathcal{M}$ and assignment σ is a function $[\![\cdot]\!]^{\sigma}_{\mathcal{M}} : \mathsf{Term} \to \mathcal{S}$ defined by:

- **Variable** $[\![v]\!]^{\sigma}_{\mathcal{M}} = \sigma(v)$
- **Function application** $[\![f(t_1, \ldots, t_n)]\!]^{\sigma}_{\mathcal{M}} = |f|([\![t_1]\!]^{\sigma}_{\mathcal{M}}, \ldots, [\![t_n]\!]^{\sigma}_{\mathcal{M}})$

See `Logos.Foundation.Assignment.Assignment`.

2.6 Verification and Falsification

The *verification* ($\Vdash$) and *falsification* ($\dashv\!\!\Vert$) of a formula φ is defined recursively relative to a model $\mathcal{M}$, assignment σ, and state s.

2.6.1 Atomic Formulas

Definition 2.14 (Atomic Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash F(t_1, \ldots, t_n) \ \textit{iff} \ \exists f \in |F|^+ \text{ where } s = f([\![t_1]\!]^\sigma_{\mathcal{M}}, \ldots, [\![t_n]\!]^\sigma_{\mathcal{M}}) \quad (1)$$

$$\mathcal{M}, \sigma, s \dashv\!\!\Vert F(t_1, \ldots, t_n) \ \textit{iff} \ \exists f \in |F|^- \text{ where } s = f([\![t_1]\!]^\sigma_{\mathcal{M}}, \ldots, [\![t_n]\!]^\sigma_{\mathcal{M}}) \quad (2)$$

2.6.2 Lambda Abstraction

Definition 2.15 (Lambda Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash (\lambda v.\varphi)(t) \ \textit{iff} \ \mathcal{M}, \sigma[[\![t]\!]^\sigma_{\mathcal{M}}/v], s \Vdash \varphi \quad (3)$$

$$\mathcal{M}, \sigma, s \dashv\!\!\Vert (\lambda v.\varphi)(t) \ \textit{iff} \ \mathcal{M}, \sigma[[\![t]\!]^\sigma_{\mathcal{M}}/v], s \dashv\!\!\Vert \varphi \quad (4)$$

2.6.3 Negation

Definition 2.16 (Negation Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash \neg\varphi \ \textit{iff} \ \mathcal{M}, \sigma, s \dashv\!\!\Vert \varphi \quad (5)$$

$$\mathcal{M}, \sigma, s \dashv\!\!\Vert \neg\varphi \ \textit{iff} \ \mathcal{M}, \sigma, s \Vdash \varphi \quad (6)$$

Remark 2.17. Negation exchanges verifiers and falsifiers: what makes φ true makes $\neg\varphi$ false, and vice versa. This maintains bilateral symmetry and ensures that a state wholly relevant to φ is equally relevant to $\neg\varphi$.

2.6.4 Conjunction

Definition 2.18 (Conjunction Verification and Falsification)**.**

$$\begin{aligned} \mathcal{M}, \sigma, s \Vdash \varphi \wedge \psi \ \textit{iff} \ & s = t \sqcup u \text{ for some } t, u \text{ where } \mathcal{M}, \sigma, t \Vdash \varphi \\ & \text{and } \mathcal{M}, \sigma, u \Vdash \psi \quad (7) \\ \mathcal{M}, \sigma, s \dashv\!\!\Vert \varphi \wedge \psi \ \textit{iff} \ & \mathcal{M}, \sigma, s \dashv\!\!\Vert \varphi, \text{ or } \mathcal{M}, \sigma, s \dashv\!\!\Vert \psi, \text{ or } s = t \sqcup u \text{ for} \\ & \text{some } t, u \text{ where } \mathcal{M}, \sigma, t \dashv\!\!\Vert \varphi \text{ and } \mathcal{M}, \sigma, u \dashv\!\!\Vert \psi \quad (8) \end{aligned}$$

Remark 2.19. A conjunction $\varphi \wedge \psi$ is verified by a state that is the *fusion* of a verifier for φ and a verifier for ψ. If state t verifies "Alice is thinking" and state u verifies "Alice is writing," then $t \sqcup u$ verifies "Alice is thinking and

writing." Crucially, $t \sqcup u$ does *not* verify "Alice is thinking" alone since u is irrelevant to that claim. The non-monotonicity of the verification and falsification relations is characteristic of exact semantics.

The falsification clause is *inclusive*: a conjunction is falsified by anything that falsifies either conjunct, or by a fusion of falsifiers for both conjuncts. This ensures that $\varphi \wedge \varphi$ has the same falsifiers as φ so that $\varphi \wedge \varphi \equiv \varphi$ comes out valid, providing a natural idempotence law.

2.6.5 Disjunction

Definition 2.20 (Disjunction Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash \varphi \vee \psi \ \textit{iff}\ \mathcal{M}, \sigma, s \Vdash \varphi, \text{ or } \mathcal{M}, \sigma, s \Vdash \psi, \text{ or } s = t \sqcup u \text{ for some } t, u \text{ where } \mathcal{M}, \sigma, t \Vdash \varphi \text{ and } \mathcal{M}, \sigma, u \Vdash \psi \tag{9}$$

$$\mathcal{M}, \sigma, s \dashv\!\Vdash \varphi \vee \psi \ \textit{iff}\ s = t \sqcup u \text{ for some } t, u \text{ where } \mathcal{M}, \sigma, t \dashv\!\Vdash \varphi \text{ and } \mathcal{M}, \sigma, u \dashv\!\Vdash \psi \tag{10}$$

Remark 2.21 (Disjunction Semantics). Disjunction is dual to conjunction: verification and falsification swap roles. A disjunction $\varphi \vee \psi$ is falsified by a fusion of falsifiers for each disjunct—the state must be wholly relevant to falsifying both. The verification clause is inclusive, mirroring the falsification clause for conjunction, making $\varphi \vee \varphi \equiv \varphi$ valid.

2.6.6 Top and Bottom

Definition 2.22 (Top and Bottom Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash \top \text{ for all } s \in \mathcal{S} \tag{11}$$

$$\mathcal{M}, \sigma, s \dashv\!\Vdash \top \ \textit{iff}\ s = \blacksquare \ (\textit{full state}) \tag{12}$$

$$\mathcal{M}, \sigma, s \nVdash \bot \text{ for all } s \tag{13}$$

$$\mathcal{M}, \sigma, s \dashv\!\Vdash \bot \ \textit{iff}\ s = \square \ (\textit{null state}) \tag{14}$$

2.6.7 Propositional Identity

Definition 2.23 (Propositional Identity Verification and Falsification)**.**

$$\begin{aligned}
\mathcal{M}, \sigma, s \Vdash \varphi \equiv \psi \ \textit{iff}\ & s = \square \text{ and for all } t \text{ both:} \\
& (1)\ \mathcal{M}, \sigma, t \Vdash \varphi \text{ just in case } \mathcal{M}, \sigma, t \Vdash \psi; \text{ and} \\
& (2)\ \mathcal{M}, \sigma, t \dashv\!\Vert\ \varphi \text{ just in case } \mathcal{M}, \sigma, t \dashv\!\Vert\ \psi \qquad (15) \\
\mathcal{M}, \sigma, s \dashv\!\Vert\ \varphi \equiv \psi \ \textit{iff}\ & \text{at least one of the following:} \\
& (1)\ \mathcal{M}, \sigma, s \Vdash \varphi \text{ and } \mathcal{M}, \sigma, s \nVdash \psi; \\
& (2)\ \mathcal{M}, \sigma, s \nVdash \varphi \text{ and } \mathcal{M}, \sigma, s \Vdash \psi; \\
& (3)\ \mathcal{M}, \sigma, s \dashv\!\Vert\ \varphi \text{ and } \mathcal{M}, \sigma, s \not\dashv\!\Vert\ \psi; \text{ or} \\
& (4)\ \mathcal{M}, \sigma, s \not\dashv\!\Vert\ \varphi \text{ and } \mathcal{M}, \sigma, s \dashv\!\Vert\ \psi. \qquad (16)
\end{aligned}$$

See `Logos.Foundation.Semantics.verifies` for the implementation.

2.7 Essence and Ground

These constitutive relations can be defined via propositional identity:

Definition 2.24 (Essence and Ground)**.**

$$\varphi \sqsubseteq \psi := \varphi \wedge \psi \equiv \psi \qquad \text{“}\varphi \text{ is essential to } \psi\text{” (conjunctive part)} \qquad (17)$$

$$\varphi \leqslant \psi := \varphi \vee \psi \equiv \psi \qquad \text{“}\varphi \text{ grounds } \psi\text{” (disjunctive part)} \qquad (18)$$

Definition 2.25 (Reduction)**.** *Reduction* combines essence and ground:

$$\varphi \Rightarrow \psi := (\varphi \sqsubseteq \psi) \wedge (\varphi \leqslant \psi)$$

Reduction expresses that φ is both essential to (necessary for) and grounds (sufficient for) ψ. Examples include $\varphi \Rightarrow \varphi \vee (\varphi \wedge \psi)$ and $\varphi \Rightarrow \varphi \wedge (\varphi \vee \psi)$.

See `Logos.Foundation.Relations.essence`.

2.8 Bilateral Propositions

Definition 2.26 (Bilateral Proposition)**.** A *bilateral proposition* $P : \mathbb{P}$ is an ordered pair $\langle V, F \rangle$ where V and F are subsets of $\mathcal{S}$ closed under fusion:

- **Fusion closure for verifiers**: $s \sqcup t \in V$ for any $s, t \in V$
- **Fusion closure for falsifiers**: $s \sqcup t \in F$ for any $s, t \in F$

Definition 2.27 (Propositional Operations)**.** For sets of states $X, Y \subseteq \mathcal{S}$ and bilateral propositions $P = \langle P^+, P^- \rangle$ and $Q = \langle Q^+, Q^- \rangle$, we may define the operations on their verifier and falsifier sets as follows:

- **Pairwise Fusion**: $X \otimes Y := \{s \sqcup t : s \in X, t \in Y\}$
- **Closed Fusion**: $X \oplus Y := X \cup Y \cup (X \otimes Y)$
- **Negation**: $\neg P := \langle P^-, P^+ \rangle$
- **Conjunction**: $P \wedge Q := \langle P^+ \otimes Q^+, P^- \oplus Q^- \rangle$
- **Disjunction**: $P \vee Q := \langle P^+ \oplus Q^+, P^- \otimes Q^- \rangle$
- **Essence**: $P \sqsubseteq Q := P \wedge Q = Q$
- **Ground**: $P \sqsubseteq Q := P \vee Q = Q$

Definition 2.28 (Bilattice)**.** A *bilattice* is a structure $B = \langle \mathbb{P}, \sqsubseteq, \leqslant, \neg \rangle$ where:

- $\langle \mathbb{P}, \sqsubseteq \rangle$ is a complete lattice (the *essence* order)
- $\langle \mathbb{P}, \leqslant \rangle$ is a complete lattice (the *ground* order)
- $\neg : \mathbb{P} \to \mathbb{P}$ is an *involutive negation*: $\neg\neg P = P$
- Negation exchange: $P \sqsubseteq Q \leftrightarrow \neg P \leqslant \neg Q$ and $P \leqslant Q \leftrightarrow \neg P \sqsubseteq \neg Q$.

A bilattice is *interlaced* when its orders interact uniformly: $(P \star R) \circ (Q \star R)$ whenever $P \circ Q$, for any lattice operation $\star$ and either order $\circ$.

Remark 2.29. The essence order $P \sqsubseteq Q$ captures a constitutive explanatory reading of "necessary for" whereby: (1) every verifier for Q has a part that verifies P; (2) any fusion of a verifier for P together with a verifier for Q also verifies Q; and (3) every falsifier for P also falsifies Q. Intuitively, P being the case is *part of what it is for* Q to be the case.

The ground order $P \leqslant Q$ captures a constitutive explanatory reading of "sufficient for" whereby: (1) every verifier for P also verifies Q; (2) every falsifier for Q has a part that falsifies P; and (3) any fusion of a falsifier for P together with a falsifier for Q also falsifies Q. Intuitively, P being the case *determines* that it is the case that Q.

Remark 2.30. The bilateral propositions form a *non-interlaced* bilattice:

- Conjunction is the least upper bound with respect to $\sqsubseteq$
- Disjunction is the least upper bound with respect to $\leqslant$
- The two orders need not interact uniformly (hence non-interlaced)

The failure of interlacing reflects the hyperintensionality of the semantics: giving up Boolean identities allows the two orders to diverge.

See `Logos.Foundation.Proposition.BilateralProp`.

2.9 Constitutive Consequence

Definition 2.31 (Identity Sentences)**.** An *identity sentence* is a well-formed formula of the form $\varphi \equiv \psi$ where the outermost connective is propositional identity. The *Constitutive Foundation* evaluates only identity sentences at the null state, as non-identity sentences require the contingent evaluation context provided by the *Dynamical Foundation*.

Definition 2.32 (Constitutive Consequence)**.** Let Γ be a set of identity sentences (Definition 2.31) and φ an identity sentence. Then:

$$\begin{aligned} \Gamma \vDash \varphi \ \textit{iff} \ & \text{for any model } \mathcal{M} \text{ and assignment } \sigma : \\ & \text{if } \mathcal{M}, \sigma, \square \vDash \psi \text{ for all } \psi \in \Gamma, \text{ then } \mathcal{M}, \sigma, \square \vDash \varphi \end{aligned}$$

Remark 2.33 (Identity and Contingency). The *Constitutive Foundation* does not provide the semantic resources to evaluate non-identity sentences which depend on contingent states of affairs rather than purely structural relations in state space. The *Dynamical Foundation* encodes contingencies by including a world-history, time, and list of temporal references in addition to a variable assignment in the contexts at which sentences are evaluated.

3 Dynamical Foundation

The *Dynamical Foundation* extends the *Constitutive Foundation* with both temporal and modal structure by including a task relation parameterized by durations, enabling the evaluation of truth relative to a world-history and time. While the hyperintensional foundation provide fine-grained semantic contents, specifying verifiers and falsifiers for sentences does not suffice to model contingency. This extension adds an intensional layer by constructing possible states, state evolutions, world-states, and world-histories in order to assign truth-values to contingent and temporary sentences.

3.1 Additional Syntactic Primitives

The *Dynamical Foundation* interprets the following additional syntactic primitives beyond those of the *Constitutive Foundation*:

- **Actuality Predicate**: Act ("t is actual")
- **Quantification**: $\forall$ ("for all")
- **Counterfactual conditional**: $\square\!\!\rightarrow$ ("if φ were the case, then χ would be the case")

- **Temporal operators**: H ("it has always been that"), G ("it is always going to be that")
- **Extended temporal operators**: $\lhd$ ("φ since ψ" means ψ was true at some past time and φ has been true ever since), $\rhd$ ("φ until ψ" means ψ will be true at some future time and φ is true until then)
- **Stability operator**: Stab ("stably φ")
- **Store/recall operators**: $\uparrow^i$ ("store current time"), $\downarrow^i$ ("recall stored time")

3.2 Well-Formed Sentences

Definition 3.1 (Well-Formed Formulas)**.** The set of *well-formed formulas* extends the Constitutive Foundation:

$$\varphi, \psi ::= \mathsf{Act}(t) \mid \Box\varphi \mid H\varphi \mid G\varphi \mid \varphi \lhd \psi \mid \varphi \rhd \psi \mid \varphi_{\text{CF}} \mathrel{\Box\!\!\rightarrow} \psi \mid \uparrow^i \varphi \mid \downarrow^i \varphi$$

where φ_{CF} ranges over Constitutive Foundation formulas.

3.3 Derived Operators

Definition 3.2 (Derived Operators)**.** The following operators are defined in terms of primitives:

$$\Diamond\varphi := \neg\Box\neg\varphi \qquad \text{"Possibly } \varphi\text{"} \tag{19}$$

$$P\varphi := \neg H\neg\varphi \qquad \text{"Sometimes past } \varphi\text{"} \tag{20}$$

$$F\varphi := \neg G\neg\varphi \qquad \text{"Sometimes future } \varphi\text{"} \tag{21}$$

$$\triangle\varphi := H\varphi \wedge \varphi \wedge G\varphi \qquad \text{"Always } \varphi\text{"} \tag{22}$$

$$\triangledown\varphi := P\varphi \vee \varphi \vee F\varphi \qquad \text{"Sometimes } \varphi\text{"} \tag{23}$$

$$\Box\varphi := \top \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{"Necessarily } \varphi\text{"} \tag{24}$$

$$\Diamond\!\!\rightarrow \varphi\psi := \neg(\varphi \mathrel{\Box\!\!\rightarrow} \neg\psi) \qquad \text{"}\varphi \text{ might-counterfactual } \psi\text{"} \tag{25}$$

Definition 3.3 (Derived Operators)**.** The following notational conventions ease exposition where $\lambda v.\varphi$ binds the variable v in φ and $\forall$ is the universal quantifier applied to the resulting property, where $\exists v.\varphi$ is similar.

1. **Universal Quantification**: $\forall v.\varphi := \forall(\lambda v.\varphi)$
2. **Existential Quantification**: $\exists v.\varphi := \neg\forall v.\neg\varphi$

These abbreviations reduce quantified formulas to applications of the universal quantifier $\forall$ (or its dual $\exists$) to lambda-abstracted properties.

See `Logos.Dynamics.Syntax.DynamicalFormula`.

3.4 Operator Summary

Category	Symbol	Reading
Modal	$\Box\varphi$	"Necessarily φ"
	$\Diamond\varphi$	"Possibly φ"
Past Temporal	$H\varphi$	"It has always been that φ"
	$P\varphi$	"It was the case that φ"
Future Temporal	$G\varphi$	"It will always be that φ"
	$F\varphi$	"It will be the case that φ"
Extended	$\varphi \lhd \psi$	"φ since ψ"
	$\varphi \rhd \psi$	"φ until ψ"
Counterfactual	$\varphi \mathrel{\Box\!\!\rightarrow} \chi$	"If φ were the case, then χ would be the case"
	$\varphi \mathrel{\Diamond\!\!\rightarrow} \chi$	"If φ were the case, then χ might be the case"
Store/Recall	$\uparrow^i \varphi$	"Store current time and evaluate φ"
	$\downarrow^i \varphi$	"Recall stored time and evaluate φ"

Table 1: Dynamical Foundation Operators

3.5 Dynamical Frame

Definition 3.4. A *dynamical frame* is a structure $\mathcal{F} = \langle \mathcal{S}, \sqsubseteq, \mathcal{D}, \Rightarrow \rangle$ where:

- $\langle \mathcal{S}, \sqsubseteq \rangle$ is a constitutive frame
- $\mathcal{D} = \langle D, +, \leqslant \rangle$ is a totally ordered abelian group
- $\Rightarrow$ is a ternary relation on $\mathcal{S} \times D \times \mathcal{S}$ satisfying the compositionality constraint that $s \Rightarrow_{x+y} r$ whenever $s \Rightarrow_x t$ and $t \Rightarrow_y r$.

Notation 3.5. We write $s \Rightarrow_d t$ to indicate that there is a *task* from s to t in duration d where a task is a possible state transition.

3.6 State Modality Definitions

Definition 3.6 (State Modality)**.** Let $\mathcal{F} = \langle \mathcal{S}, \sqsubseteq, \mathcal{D}, \Rightarrow \rangle$ be a dynamical frame. We define the following predicates on states:

$$\mathsf{Possible}(s) := s \Rightarrow_0 s \tag{26}$$

$$\mathsf{Compatible}(s,t) := \mathsf{Possible}(s \sqcup t) \tag{27}$$

$$\mathsf{Maximal}(s) := \forall t : \mathcal{S}.\ \mathsf{Compatible}(t,s) \to t \sqsubseteq s \tag{28}$$

$$\mathsf{WorldState}(w) := \mathsf{Possible}(w) \wedge \mathsf{Maximal}(w) \tag{29}$$

$$s \sim t := \exists d : D.\ s \Rightarrow_d t \vee t \Rightarrow_d s \tag{30}$$

$$\mathsf{Necessary}(s) := \forall t : \mathcal{S}.\ s \sim t \to s = t \tag{31}$$

Remark 3.7. A world-state w satisfying $\mathsf{WorldState}(w)$ is maximal in the sense that every compatible state is a part of w.

Definition 3.8 (Maximal Compatible Part)**.** For states $r, t : \mathcal{S}$, the set of *maximal t-compatible parts* of r is:

$$[r]_t := \{s : \mathcal{S} \mid s \sqsubseteq r \wedge \mathsf{Compatible}(s,t) \wedge \forall s'.\ (s \sqsubseteq s' \sqsubseteq r \wedge \mathsf{Compatible}(s',t)) \to s' \sqsubseteq s\}$$

3.7 Task Relation Constraints

Definition 3.9. The task relation $\Rightarrow$ satisfies:

- **L-Parthood**: If $d \sqsubseteq s$ and $s \Rightarrow_x t$, then $d \Rightarrow_x r$ for some $r \sqsubseteq t$
- **R-Parthood**: If $r \sqsubseteq t$ and $s \Rightarrow_x t$, then $d \Rightarrow_x r$ for some $d \sqsubseteq s$
- **L-Contained**: If $s : P$, $d \sqsubseteq s$, and $d \Rightarrow_x r$, then $s \Rightarrow_x t \sqcup r$ for some $t : \mathcal{S}$
- **R-Contained**: If $t : P$, $r \sqsubseteq t$, and $d \Rightarrow_x r$, then $s \sqcup d \Rightarrow_x t$ for some $s : \mathcal{S}$
- **Maximal**: If $s : \mathcal{S}$ and $t : P$, there is a maximal t-compatible part $r : [s]_t$

Remark 3.10. The *Containment* constraints ensure that tasks between parts of possible states can be extended to tasks between the states themselves. The *Maximality* constraint ensures that there are no strictly increasing series $r_1 \sqsubset r_2 \sqsubset \ldots \sqsubset s$ of parts $r_i \sqsubseteq s$ compatible with t with no upper bound.

See `Logos.Dynamics.Frame.DynamicalFrame`.

3.8 World-History

Definition 3.11 (Evolution)**.** An *evolution* over a dynamical frame $\mathcal{F}$ is a function $\tau : X \to \mathcal{S}$ where:

- $X \subseteq D$ is a convex subset of the temporal order

- $\tau(x) \Rightarrow_{y-x} \tau(y)$ for all times $x, y \in X$ where $x \leqslant y$

Definition 3.12 (Evolution Parthood)**.** For evolutions $\tau, \sigma : X \to \mathcal{S}$, we define $\tau \sqsubseteq \sigma$ iff $\tau(x) \sqsubseteq \sigma(x)$ for all $x \in X$.

Definition 3.13 (Possible and Maximal Evolution)**.** An evolution τ is:

- *possible* iff $\mathsf{Possible}(\tau(x))$ for all x in its domain
- *maximal* iff for any possible evolution σ where $\tau \sqsubseteq \sigma$, we have $\sigma = \tau$

Definition 3.14 (World-History)**.** A *world-history* is a maximal possible evolution $\tau : X \to \mathcal{S}$.

Theorem 3.15 (Containment)**.** *An evolution τ is a world-history iff* $\mathsf{WorldState}(\tau(x))$ *for all x in its domain.*

Notation 3.16. The type of all world-histories over $\mathcal{F}$ is denoted $\mathsf{WorldHistory}(\mathcal{F})$.

3.9 Dynamical Model

Definition 3.17 (Dynamical Model)**.** A *dynamical model* is any $\mathcal{M} = \langle \mathcal{S}, \sqsubseteq, \mathcal{D}, \Rightarrow, |\cdot| \rangle$ where $\langle \mathcal{S}, \sqsubseteq, \mathcal{D}, \Rightarrow \rangle$ is a dynamical frame and:

- $|\cdot|$ is an interpretation defined in the *Constitutive Foundation*
-
-

See `Logos.Dynamics.Model.DynamicalModel`.

3.9.1 Universal Quantification

Definition 3.18 (Universal Quantification Verification and Falsification)**.**

$$\begin{aligned} \mathcal{M}, \sigma, s \Vdash \forall \varphi(v) \ \textit{iff} \ & (1)\ \mathcal{M}, \sigma[d/v], d \Vdash \varphi \text{ for all } d \sqsubseteq s; \text{ and} \\ & (2)\ r \sqsubseteq s \text{ for all } r \circ s \end{aligned} \tag{32}$$

$$\begin{aligned} \mathcal{M}, \sigma, s \dashv\!\Vert \forall \varphi(v) \ \textit{iff} \ & (1)\ \mathcal{M}, \sigma[d/v], d \dashv\!\Vert \varphi \text{ for some } d \sqsubseteq s; \text{ and} \\ & (2)\ r \sqsubseteq s \text{ for all } r \circ s \end{aligned} \tag{33}$$

3.10 Truth Conditions

Truth is evaluated relative to five parameters:

- $\mathcal{M}$: a dynamical model
- τ: a world-history (an evolution $\tau : X \to \mathcal{S}$ over convex $X \subseteq D$)

- x: a time point ($x \in D$ such that $x \in \mathrm{dom}(\tau)$)
- σ: a variable assignment ($\sigma : \mathsf{Var} \to \mathcal{S}$)
- $\vec{\imath}$: a temporal index ($\vec{\imath} = \langle i_1, i_2, \ldots \rangle$ storing times for recall operators)

We write $\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi$ to indicate that φ is true at $\mathcal{M}$, τ, x, σ, $\vec{\imath}$.

3.10.1 Atomic Sentences

Definition 3.19 (Atomic Truth)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models F(t_1, \ldots, t_n) \;\; \textit{iff} \;\; \exists f \in |F|^+ : f([\![t_1]\!]^\sigma_{\mathcal{M}}, \ldots, [\![t_n]\!]^\sigma_{\mathcal{M}}) \sqsubseteq \tau(x) \quad (34)$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \dashv F(t_1, \ldots, t_n) \;\; \textit{iff} \;\; \exists f \in |F|^- : f([\![t_1]\!]^\sigma_{\mathcal{M}}, \ldots, [\![t_n]\!]^\sigma_{\mathcal{M}}) \sqsubseteq \tau(x) \quad (35)$$

Theorem 3.20 (Bivalence for World-Histories)**.** *For any world-history* τ : $\mathsf{WorldHistory}(\mathcal{F})$*, model* $\mathcal{M}$*, time* $x : D$*, assignment* σ*, and temporal index* $\vec{\imath}$*:*

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \;\; \textit{iff} \;\; \mathcal{M}, \tau, x, \sigma, \vec{\imath} \not\dashv \varphi$$

Remark 3.21. This bivalence property breaks down for non-maximal evolutions τ that are not world-histories.

3.10.2 Lambda Abstraction and Quantification

Definition 3.22 (Lambda Application and Universal Quantification)**.** The primitive semantic clauses for lambda application and universal quantification are:

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models (\lambda v.\varphi)(t) \;\; \textit{iff} \;\; \mathcal{M}, \tau, x, \sigma[[\![t]\!]^\sigma_{\mathcal{M}}/v], \vec{\imath} \models \varphi \quad (36)$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \forall\varphi(v) \;\; \textit{iff} \;\; \mathcal{M}, \tau, x, \sigma[s/v], \vec{\imath} \models \varphi(v) \text{ for all } s : \mathcal{S} \quad (37)$$

where $\varphi(v)$ is a property (a formula with at least one free variable v), and $\sigma[s/v]$ extends the assignment σ by mapping v to s.

Remark 3.23 (Reduction of Quantified Formulas). To evaluate $\forall v.\varphi$, we expand via the syntactic convention and then apply the primitive semantic clauses:

$$\begin{aligned} \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \forall v.\varphi \;\; &\textit{iff} \;\; \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \forall(\lambda v.\varphi) && \text{(syntactic sugar)} \\ &\textit{iff} \;\; \mathcal{M}, \tau, x, \sigma[s/v], \vec{\imath} \models \varphi \text{ for all } s : \mathcal{S} && \text{(by } \forall \text{ and } \lambda \text{ semantics)} \end{aligned}$$

This matches the Lean implementation where `Formula.all` and `Formula.lambdaApp` are separate constructors.

Remark 3.24 (Naming Conventions). In this LaTeX presentation, we distinguish:

- **Metalanguage times**: denoted $x, y, z \in D$
- **Object language variables**: denoted $v, v_1, v_2, \ldots$
- **States**: denoted $s, t, r \in \mathcal{S}$

In the Lean implementation (`Truth.lean`):

- Times use identifiers `t`, `y`
- Variables use `x` (of type `String`)
- States use identifiers `s`, `t`, `r`

This notational divergence is intentional to avoid confusion between metalanguage time variables (ranging over the temporal order D) and object language term variables (bound by quantifiers and lambda abstractions).

3.10.3 Actuality Predicate

Definition 3.25 (Actuality)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \mathsf{Act}(t) \ \textit{iff} \ [\![t]\!]^{\sigma}_{\mathcal{M}} \sqsubseteq \tau(x)$$

Remark 3.26 (Restricted Quantification). The actuality predicate enables restricting quantification to actually existing objects:

$$\forall y(\mathsf{Act}(y) \to \varphi) \text{ is true iff } \varphi \text{ holds for all states that are parts of } \tau(x)$$

The unrestricted universal quantifier validates the Barcan formulas ($\forall x \Box \varphi \to \Box \forall x \varphi$ and its converse), while the actuality-restricted quantifier does not.

Additional restricted quantifiers can be defined:

- *Sometimes actual*: $\forall y(\bigtriangledown \mathsf{Act}(y) \to \varphi)$ quantifies over all states that are sometimes actual
- *Possibly actual*: $\forall y(\Diamond \mathsf{Act}(y) \to \varphi)$ quantifies over all states that are possibly actual

Both are restrictions of the full unrestricted quantifier, with the unrestricted and possibly actual quantifiers validating the Barcan formulas.

3.10.4 Extensional Connectives

Definition 3.27 (Extensional Connectives)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \neg\varphi \ \textit{iff} \ \mathcal{M}, \tau, x, \sigma, \vec{\imath} \dashv \varphi \tag{38}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \wedge \psi \ \textit{iff} \ \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \text{ and } \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \psi \tag{39}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \vee \psi \ \textit{iff} \ \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \text{ or } \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \psi \tag{40}$$

3.10.5 Stability Operator (Primitive)

Definition 3.28 (Stability)**.** The *stability operator* $\mathsf{Stab}\varphi$ quantifies over all histories sharing the current world-state:

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \mathsf{Stab}\varphi \ \textit{iff} \ \mathcal{M}, \alpha, x, \sigma, \vec{\imath} \models \varphi \text{ for all } \alpha : \mathsf{WorldHistory}(\mathcal{F}) \text{ where } \alpha(x) = \tau(x)$$

Remark 3.29. The stability operator cannot be defined in terms of the counterfactual conditional. It captures what is true at the current world-state across all compatible histories.

3.10.6 Modal Operators (Derived)

Remark 3.30 (Modal Operators as Derived). The modal operators are derived from the counterfactual:

- $\Box\varphi := \top \mathrel{\Box\!\!\rightarrow} \varphi$ (necessity)
- $\Diamond\varphi := \neg\Box\neg\varphi$ (possibility)

The semantic clauses follow from these definitions:

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \Box\varphi \ \textit{iff} \ \mathcal{M}, \alpha, x, \sigma, \vec{\imath} \models \varphi \text{ for all } \alpha : \mathsf{WorldHistory}(\mathcal{F}) \tag{41}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \Diamond\varphi \ \textit{iff} \ \mathcal{M}, \alpha, x, \sigma, \vec{\imath} \models \varphi \text{ for some } \alpha : \mathsf{WorldHistory}(\mathcal{F}) \tag{42}$$

This yields an S5 modal logic.

3.10.7 Dynamical Tense Operators

Definition 3.31 (Primitive Tense Operators)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models H\varphi \ \textit{iff} \ \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for all } y : D \text{ where } y < x \tag{43}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models G\varphi \ \textit{iff} \ \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for all } y : D \text{ where } y > x \tag{44}$$

Remark 3.32 (Derived Tense Operators). The "sometimes" operators are derived:

- $P\varphi := \neg H \neg \varphi$
- $F\varphi := \neg G \neg \varphi$

Their semantic clauses follow:

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models P\varphi \ \textit{iff} \ \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for some } y : D \text{ where } y < x \tag{45}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models F\varphi \ \textit{iff} \ \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for some } y : D \text{ where } y > x \tag{46}$$

3.10.8 Extended Tense Operators

Definition 3.33 (Since and Until)**.**

$$\begin{aligned} \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \lhd \psi \ \textit{iff} \ & \text{there exists } z < x \text{ where } \mathcal{M}, \tau, z, \sigma, \vec{\imath} \models \psi \text{ and} \\ & \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for all } y \text{ where } z < y < x \end{aligned} \tag{47}$$

$$\begin{aligned} \mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \rhd \psi \ \textit{iff} \ & \text{there exists } z > x \text{ where } \mathcal{M}, \tau, z, \sigma, \vec{\imath} \models \psi \text{ and} \\ & \mathcal{M}, \tau, y, \sigma, \vec{\imath} \models \varphi \text{ for all } y \text{ where } x < y < z \end{aligned} \tag{48}$$

3.10.9 Counterfactual Conditional

Definition 3.34 (Counterfactual Conditional)**.**

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \varphi \mathrel{\Box\!\!\rightarrow} \chi \ \textit{iff} \ \mathcal{M}, \beta, x, \sigma, \vec{\imath} \models \chi \text{ whenever } \mathcal{M}, \sigma, s \Vdash \varphi \text{ and } \beta : \langle \tau \rangle_s^x$$

Definition 3.35 (Imposition)**.** We write $t \rightarrow_w w'$ ("imposing t on w yields w'") iff there exists a maximal t-compatible part $s \in [w]_t$ where $s \sqcup t \sqsubseteq w'$.

Definition 3.36 (Alternative Worlds)**.** For a world-history τ, state $s : \mathcal{S}$, and time $x : D$, the *alternative worlds* are:

$$\langle \tau \rangle_s^x := \{\beta : \mathsf{WorldHistory}(\mathcal{F}) \mid s \rightarrow_{\tau(x)} \beta(x)\}$$

where $t \rightarrow_w w'$ holds iff there exists $r \in [w]_t$ such that $r \sqcup t \sqsubseteq w'$.

3.10.10 Store and Recall Operators

Definition 3.37 (Store and Recall)**.** For cross-temporal reference, the context includes a temporal index $\vec{\imath} = \langle i_1, i_2, \ldots \rangle$ of stored times:

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \uparrow^k \varphi \ \textit{iff} \ \mathcal{M}, \tau, x, \sigma, \vec{\imath}[x/i_k] \models \varphi \tag{49}$$

$$\mathcal{M}, \tau, x, \sigma, \vec{\imath} \models \downarrow^k \varphi \ \textit{iff} \ \mathcal{M}, \tau, i_k, \sigma, \vec{\imath} \models \varphi \tag{50}$$

where $\vec{\imath}[x/i_k]$ replaces i_k with current time x.

See `Logos.Dynamics.Semantics.Satisfies`

3.11 Bimodal Interaction Principles

The task semantics validates these perpetuity principles connecting modal and temporal operators:

Definition 3.38 (Perpetuity Principles)**.**

P1 :	$\Box\varphi \to \triangle\varphi$	Necessary implies always	(51)
P2 :	$\triangledown\varphi \to \Diamond\varphi$	Sometimes implies possible	(52)
P3 :	$\Box\triangle\varphi \leftrightarrow \triangle\Box\varphi$	Necessary-always commutativity	(53)
P4 :	$\Diamond\triangledown\varphi \leftrightarrow \triangledown\Diamond\varphi$	Possible-sometimes commutativity	(54)
P5 :	$\Box\varphi \to \Box\triangle\varphi$	Necessary implies necessary-always	(55)
P6 :	$\triangledown\Diamond\varphi \to \Diamond\varphi$	Sometimes-possible implies possible	(56)

3.12 Temporal Frame Constraints

Different temporal structures yield different valid principles. The framework does not assume discrete time:

Constraint	Description	Corresponding Axiom
Dense	Between any two times there is another	$GG\varphi \to G\varphi$
Complete	Every bounded set has a least upper bound	$\triangle(H\varphi \to FH\varphi) \to (H\varphi \to G\varphi)$
Unbounded Past	No earliest time	$P\top$
Unbounded Future	No latest time	$F\top$

Table 2: Temporal frame constraints and corresponding axioms

3.13 Logical Consequence

Definition 3.39 (Logical Consequence)**.**

$\Gamma \vDash \varphi$ *iff* for any $\mathcal{M}, \tau \in H_{\mathcal{F}}, x \in D, \sigma, \vec{\imath}$: if $\mathcal{M}, \tau, x, \sigma, \vec{\imath} \vDash \psi$ for all $\psi \in \Gamma$, then $\mathcal{M}, \tau, x, \sigma, \vec{\imath} \vDash \varphi$

Remark 3.40. Unlike the semantics presented in te *Constitutive Foundation*, the semantics for the *Dynamics Foundation* evaluates truth relative to world-histories and times, fixing the contingencies required to assign truth-values rather than verifiers and falsifiers alone. Nevertheless, the hyperintensional distinctions between necessarily equivalent sentences are still preserved since propositions still differ if their exact verifiers differ.

3.14 Counterfactual Logic

This section presents the axiom system for the counterfactual logic of the *Dynamics Foundation*.

Definition 3.41 (Closure Under Deduction)**.**

$$\textbf{R1}: \quad \text{If } \Gamma \vdash \chi, \text{ then } \varphi \mathrel{\Box\!\!\rightarrow} \Gamma \vdash \varphi \mathrel{\Box\!\!\rightarrow} \chi$$

Remark 3.42. This rule states that if χ is derivable from Γ, then for any antecedent φ, the counterfactual with consequent χ is derivable from the counterfactuals with consequents from Γ.

3.15 Counterfactual Axiom Schemata

Definition 3.43 (Counterfactual Axioms)**.**

$$\textbf{C1}: \quad \varphi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Identity)} \tag{57}$$

$$\textbf{C2}: \quad \varphi, \varphi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \varphi \qquad \text{(Counterfactual Modus Ponens)} \tag{58}$$

$$\textbf{C3}: \quad \varphi \mathrel{\Box\!\!\rightarrow} \psi, \varphi \wedge \psi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \varphi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Weakened Transitivity)} \tag{59}$$

$$\textbf{C4}: \quad \varphi \vee \psi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \varphi \wedge \psi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Disjunction-Conjunction)} \tag{60}$$

$$\textbf{C5}: \quad \varphi \vee \psi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \varphi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Simplification Left)} \tag{61}$$

$$\textbf{C6}: \quad \varphi \vee \psi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \psi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Simplification Right)} \tag{62}$$

$$\textbf{C7}: \quad \varphi \mathrel{\Box\!\!\rightarrow} \varphi, \psi \mathrel{\Box\!\!\rightarrow} \varphi, \varphi \wedge \psi \mathrel{\Box\!\!\rightarrow} \varphi \vdash \varphi \vee \psi \mathrel{\Box\!\!\rightarrow} \varphi \qquad \text{(Disjunction Introduction)} \tag{63}$$

Remark 3.44 (Explanation of Counterfactual Axioms).

- **C1** (Identity): Every proposition counterfactually implies itself.
- **C2** (Modus Ponens): If φ is actually true and φ would imply φ, then φ is true.
- **C3** (Weakened Transitivity): This weaker form of transitivity avoids the problematic full transitivity principle.
- **C4-C7**: These axioms govern the interaction between counterfactuals and Boolean operations on antecedents.

3.16 Modal Axiom Schemata

Definition 3.45 (Modal Axioms)**.**

M1 :	$\top$	(Truth)	(64)
M2 :	$\neg\bot$	(Non-Contradiction)	(65)
M3 :	$\varphi \to \Box\Diamond\varphi$	(Brouwer)	(66)
M4 :	$\Box\varphi \to \Box\Box\varphi$	(S4 Transitivity)	(67)
M5 :	$\Box(\varphi \to \varphi) \vdash \varphi \mathrel{\Box\!\!\to} \varphi$	(Strict-to-Counterfactual)	(68)

Remark 3.46 (S5 Modal Logic). The combination of M3 and M4 yields an S5 modal logic, which is characterized by an equivalence relation on possible worlds. In the task semantics, this arises from the structure of world-histories: any world-history is accessible from any other at any given time.

Remark 3.47 (Strict-to-Counterfactual). Axiom M5 states that if φ is a strict consequence of φ (i.e., necessarily, if φ then φ), then φ counterfactually implies φ. This connects the modal operator with the counterfactual conditional.

3.17 Derived Theorems

Theorem 3.48 (Counterfactual Strengthening)**.** *If $\varphi \mathrel{\Box\!\!\to} \varphi$ and $\varphi \mathrel{\Box\!\!\to} \psi$, then $\varphi \mathrel{\Box\!\!\to} (\varphi \wedge \psi)$.*

Theorem 3.49 (Counterfactual Weakening)**.** *If $\varphi \mathrel{\Box\!\!\to} \varphi$ and $\varphi \vdash \psi$, then $\varphi \mathrel{\Box\!\!\to} \psi$.*

Theorem 3.50 (Necessity Introduction)**.** *$\top \mathrel{\Box\!\!\to} \varphi$ is equivalent to $\Box\varphi$.*

Theorem 3.51 (Excluded Middle)**.** *$(\varphi \mathrel{\Box\!\!\to} \varphi) \vee (\varphi \mathrel{\Box\!\!\to} \neg\varphi)$ is* not *generally valid.*

Remark 3.52. The failure of excluded middle for counterfactuals (Theorem 3.51) reflects the mereological structure: when imposing an antecedent φ on the actual world, there may be multiple ways to do so, leading to different worlds where φ or $\neg\varphi$ holds. The counterfactual is true only when the consequent holds in *all* such imposed worlds.

3.18 Soundness and Completeness

Theorem 3.53 (Soundness)**.** *If $\Gamma \vdash \varphi$ using the axiom system above, then $\Gamma \vDash \varphi$.*

> [OPEN QUESTION]: Completeness of the axiom system with respect to the mereological counterfactual semantics remains an open problem. The main challenge is showing that every consistent set of formulas has a model with the required mereological structure.

See `Logos.Dynamics.Axioms.CounterfactualAxioms` for the Lean implementation.

4 Epistemic Extension

[DETAILS PENDING DEVELOPMENT]

The Epistemic Extension extends the Explanatory Extension with structures for belief, knowledge, and probability.

4.1 Frame Extension

[DETAILS PENDING DEVELOPMENT]

The epistemic frame extends the core frame with a credence function assigning probabilities to state transitions.

> [OPEN QUESTION]: What is the exact structure of the credence function? Does it assign probabilities to individual state transitions or to sets of transitions?

4.2 Operators

Operator	Intended Reading
$B_a\varphi$	Agent a believes that φ
$K_a\varphi$	Agent a knows that φ
$\Pr(\varphi) \geqslant r$	The probability of φ is at least r

Table 3: Epistemic operators

[FULL SEMANTIC CLAUSES FOR EPISTEMIC OPERATORS PENDING SPECIFICATION]

4.3 Indicative Conditionals

[DETAILS PENDING DEVELOPMENT]

> [OPEN QUESTION]: How do indicative conditionals relate to counterfactual conditionals in the semantic framework?

5 Normative Extension

[DETAILS PENDING DEVELOPMENT]

The Normative Extension extends the Explanatory Extension with structures for obligation, permission, and value.

5.1 Frame Extension

[DETAILS PENDING DEVELOPMENT]

The normative frame extends the core frame with value orderings over states.

> [OPEN QUESTION]: How are value orderings structured? Are they complete orderings or partial orderings? Are they agent-relative?

5.2 Operators

Operator	Intended Reading
$O\varphi$	It ought to be that φ
$P\varphi$	It is permitted that φ
$\varphi \succ \psi$	φ is preferred to ψ

Table 4: Normative operators

[FULL SEMANTIC CLAUSES FOR NORMATIVE OPERATORS PENDING SPECIFICATION]

5.3 Normative Explanation

[DETAILS PENDING DEVELOPMENT]

> [OPEN QUESTION]: How does normative explanation relate to causal explanation?

6 Spatial Extension

[DETAILS PENDING DEVELOPMENT]

The Spatial Extension extends the Explanatory Extension with structures for spatial reasoning and location.

6.1 Frame Extension

[DETAILS PENDING DEVELOPMENT]

The spatial frame extends the core frame with:

- **Location space** L = set of spatial locations
- **Spatial relations**: adjacency, containment, distance

> [OPEN QUESTION]: What spatial primitives are required? Should locations be mereological (with parts) or set-theoretic?

6.2 Operators

Operator	Intended Reading
Here(φ)	φ holds at the current location
Somewhere(φ)	φ holds at some location
Everywhere(φ)	φ holds at all locations
Near(φ)	φ holds at an adjacent location

Table 5: Spatial operators

[FULL SEMANTIC CLAUSES FOR SPATIAL OPERATORS PENDING SPECIFICATION]

7 Agential Extension

[DETAILS PENDING DEVELOPMENT]

The Agential Extension requires at least one of the Epistemic, Normative, or Spatial Extensions to be loaded. It provides structures for multi-agent reasoning.

Remark 7.1 (Dependency). This extension depends on at least one middle extension (Epistemic, Normative, or Spatial) being loaded. See the extension dependency diagram in section 1.1.

7.1 Frame Extension

[DETAILS PENDING DEVELOPMENT]

> [OPEN QUESTION]: What frame extensions are required for multi-agent reasoning? Does this extension add agent indices, or agent-relative accessibility relations?

7.2 Multi-Agent Operators

[DETAILS PENDING DEVELOPMENT]

> [OPEN QUESTION]: How do individual and collective agency interact in the semantic framework?

8 Reflection Extension

The Reflection Extension enables first-person metacognitive reasoning through the *I* operator. Where the Agential Extension projects epistemic operators onto other agents, the Reflection Extension projects them onto the self, enabling perspectival distance from one's own beliefs, abilities, and goal-states.

Remark 8.1 (Dependency). This extension inherits from the Epistemic Extension in parallel with the Agential Extension. Both apply the same epistemic apparatus: Agential projects outward onto other agents, Reflection projects inward onto self. See the extension dependency diagram in section 1.1.

8.1 Core Insight

The *I* operator transforms direct modal expressions into self-aware epistemic stances. Consider the difference between:

Expression	Reading	Epistemic Subject
$\mathsf{Mu}(\varphi)$	"It must be that φ"	Implicit/absent
$\mathsf{I}(\varphi)$	"I believe that φ"	Explicit (self)

The direct expression commits the speaker without self-awareness; the self-aware expression registers the speaker as the epistemic subject. This distinction follows Kaplan's character/content framework and Lewis's centered-worlds semantics.

8.2 Frame Extension

The Reflection frame extends the Agential frame with:

Component	Description
Self-Index	Distinguished agent index $\mathsf{self} \in A$
Self-Accessibility	Reflexive accessibility relation R_{self}
Self-Model	Function $\mathsf{SM} : W \to \mathsf{SelfState}$
Commitment Register	Set $\mathsf{CR}(w)$ of self-ascribed propositions at w

The self-accessibility relation R_{self} satisfies:

- **Seriality (D)**: For all w, there exists w' such that $(w, w') \in R_{\mathsf{self}}$
- **Transitivity (4)**: Positive introspection: $\mathsf{I}(\varphi) \to \mathsf{I}(\mathsf{I}(\varphi))$
- **Euclidean (5)**: Negative introspection: $\neg\mathsf{I}(\varphi) \to \mathsf{I}(\neg\mathsf{I}(\varphi))$

8.3 Truth Conditions

Given evaluation context $(\mathcal{M}, \tau, x, \sigma, \vec{i})$, the primary I operator is defined by:

Definition 8.2 (Truth Condition for I)**.** $\mathcal{M}, \tau, x, \sigma \models \mathsf{I}(\varphi)$ if and only if:

1. For all w' accessible via R_{self} from $\tau(x)$: $\mathcal{M}, \tau[w'/x], x, \sigma \models \varphi$
2. $\varphi \in \mathsf{CR}(\tau(x))$

Remark 8.3 (Commitment Register Distinction). The second condition distinguishes explicit self-attribution from mere truth across epistemic alternatives. A proposition may hold at all worlds compatible with the agent's epistemic state without the agent having explicitly endorsed it. The Commitment Register tracks which propositions the agent has actively self-ascribed, capturing the difference between implicit epistemic commitment and explicit metacognitive awareness.

8.4 Operators

8.4.1 Self-Knowledge and Self-Belief

Operator	Symbol	Reading
Metacognitive I	$\mathsf{I}(\varphi)$	"I judge/believe that φ"
Self-Knowledge	$\mathsf{I}_K(\varphi)$	"I know that φ"
Self-Belief	$\mathsf{I}_B(\varphi)$	"I believe that φ"
Self-Uncertainty	$\mathsf{I}_?(\varphi)$	"I am uncertain whether φ"

8.4.2 Ability Introspection

Operator	Symbol	Reading
Self-Ability	$\mathsf{I}_{\mathsf{Can}}(\varphi)$	"I can bring about φ"
Ability-Knowledge	$K_{\mathsf{self}}(\mathsf{Can}_{\mathsf{self}}(\varphi))$	"I know I can do φ"
Limitation-Knowledge	$K_{\mathsf{self}}(\neg\mathsf{Can}_{\mathsf{self}}(\varphi))$	"I know I cannot do φ"

8.4.3 Goal-Distance

Operator	Symbol	Reading
Goal-Distance	$\mathsf{Dist}(G, n)$	"Goal G is n steps away"
Goal-Progress	$\mathsf{Closer}(G)$	"I am getting closer to G"
Goal-Achievable	$\mathsf{Achievable}(G)$	"Goal G is achievable"

8.4.4 Attributed Belief

Operator	Symbol	Reading
Other's Belief About Self	$B_j(\mathsf{I}(\varphi))$	"Agent j believes I believe φ"
Other's Belief About Ability	$B_j(\mathsf{I}_{\mathsf{Can}}(\varphi))$	"Agent j believes I can do φ"

8.4.5 Derived Operators

The following derived operators are defined in terms of the primary I operator:

Operator	Definition	Reading
$\mathsf{I}_K(\varphi)$	$\mathsf{I}(K(\varphi))$	"I know that I know φ"
$\mathsf{I}_B(\varphi)$	$\mathsf{I}(B(\varphi))$	"I know that I believe φ"
$\mathsf{I}_?(\varphi)$	$\mathsf{I}(\neg B(\varphi) \wedge \neg B(\neg\varphi))$	"I know I am uncertain about φ"
$\mathsf{I}_{\mathsf{Can}}(\varphi)$	$\exists\vec{a}\,\mathsf{Achievable}_{\mathsf{self}}(\vec{a}) \wedge \mathsf{LeadsTo}(\vec{a}, \varphi)$	"I can bring about φ"

Remark 8.4 (Self-Knowledge vs. Self-Belief). The operators I_K and I_B differ in their factivity requirements. $\mathsf{I}_K(\varphi)$ requires that the inner knowledge claim $K(\varphi)$ be factive (hence φ is true), while $\mathsf{I}_B(\varphi)$ requires only that the agent believes φ. Self-uncertainty $\mathsf{I}_?(\varphi)$ captures metacognitive awareness of one's own agnosticism.

8.5 Key Axioms

Axiom	Name	Schema
I-T	Veridicality	$\mathsf{I}_K(\varphi) \to \varphi$
I-4	Positive Introspection	$\mathsf{I}(\varphi) \to \mathsf{I}(\mathsf{I}(\varphi))$
I-5	Negative Introspection	$\neg\mathsf{I}(\varphi) \to \mathsf{I}(\neg\mathsf{I}(\varphi))$
I-D	Consistency	$\neg\mathsf{I}(\bot)$
I-Commit	Commitment Closure	$\mathsf{I}(\varphi) \to \mathsf{Register}(\varphi)$

8.6 Temporal Interaction

The metacognitive I operator interacts non-trivially with temporal operators. The following formulas are semantically distinct:

Formula	Reading
$P(\mathsf{I}(\varphi))$	"I previously believed φ"
$\mathsf{I}(P(\varphi))$	"I believe φ was the case"
$F(\mathsf{I}(\varphi))$	"I will believe φ"
$\mathsf{I}(F(\varphi))$	"I believe φ will be the case"

Remark 8.5 (Non-Equivalence). The pairs $P(\mathsf{I}(\varphi))$ and $\mathsf{I}(P(\varphi))$ are not equivalent. The former describes a past epistemic state (what was believed then), while the latter describes a current belief about the past (what is believed now about then). An agent may currently believe something was true without having believed it at the time, and may have believed something without currently believing it was true. This demonstrates the interaction between temporal and metacognitive perspectives.

8.7 Relationship to Agential Extension

The Reflection Extension inherits from the Epistemic Extension in parallel with the Agential Extension:

$$\begin{tikzcd}[column sep=small] & \text{Epistemic Extension} \arrow[dl, "\text{outward}"'] \arrow[dr, "\text{inward}"] & \\ \text{Agential} & & \text{Reflection} \end{tikzcd}$$

Both extensions apply the epistemic apparatus in different directions:

- **Agential**: $B_j(\varphi)$ – "Agent j believes φ" (modeling others)
- **Reflection**: $\mathsf{I}(\varphi)$ – "I believe φ" (modeling self)

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